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## End Semester Examination 2025

Course: B.Tech. (Mechanical Engineering/ Aerospace Engineering)

Semester: 3rd

Paper Name: Engineering Mathematics-III

Subject Code: TMA - 303

Time: 3 Hours

Maximum Marks: 100

Note: -

- (i) All questions are compulsory.
- (ii) Answer any two sub questions among a, b & c in each main question.
- (iii) The total marks in each main question are twenty.
- (iv) Each sub question carries 10 marks.

**Q 1**

**20 Marks**

(a) Solve  $xzp + yzq = xy$

**CO1**

(b) Solve  $4 \frac{\partial^2 z}{\partial x^2} - 4 \frac{\partial^2 z}{\partial x \partial y} + \frac{\partial^2 z}{\partial y^2} = 16 \log(x + 2y)$ .

**CO1**

(c) Solve  $\frac{\partial^2 z}{\partial x^2} - 2 \frac{\partial^2 z}{\partial x \partial y} + \frac{\partial^2 z}{\partial y^2} = \sin(x + y)$ .

**CO1**

**Q 2**

**20 Marks**

(a) Using Bisection method to find a real root of an equation  $x \log_{10} x = 1.2$  correct to three decimals.

**CO2**

(b) Solve  $\frac{dy}{dx} = -xy$  numerically from  $x = 0$  to  $x = 1$  using Euler method with initial condition of  $y(0) = 1$  and step size (h) of 0.2. Find the value of  $y(1)$ . (Correct to three decimal places).

**CO2**

(c) A train is moving at the speed of 30 m/sec. Suddenly brakes are applied. The speed of the train per second after 't' seconds is given below:

|           |    |    |    |    |    |    |    |    |    |    |
|-----------|----|----|----|----|----|----|----|----|----|----|
| Time (t)  | 0  | 5  | 10 | 15 | 20 | 25 | 30 | 35 | 40 | 45 |
| Speed (v) | 30 | 24 | 19 | 16 | 13 | 11 | 10 | 8  | 7  | 5  |

Using Simpson's  $\frac{3}{8}$  rule to compute the distance moved by the train in 45 seconds.

**Q 3****20 Marks**

(a) Evaluate the Laplace transform of  $F(t) = \frac{e^{at} - \cos bt}{t}$ .

**CO3**

(b) Use Convolution theorem to find the inverse Laplace transform of the given function

$$f(s) = \frac{s}{(s^2+4)^2}.$$

**CO3**

(c) To show that  $\int_0^\infty t e^{-3t} \sin t \, dt = \frac{3}{50}$ .

**CO3****Q 4****20 Marks**

(a) If the probability density function of a continuous random variable X is given

$$\text{by } f(x) = \begin{cases} x^3 & \text{if } 0 < x < 1 \\ 0 & \text{otherwise} \end{cases}$$

Determine its mean, variance, and standard deviation.

**CO4**

(b) Using Newton-Raphson method to compute a real root of the equation

$$x^3 - x - 1 = 0 \text{ (Correct to three decimal places).}$$

**CO2**

(c) The number of airplane landings at an airport in a minute (X) and their probabilities are given by

|          |      |      |      |      |      |      |      |
|----------|------|------|------|------|------|------|------|
| $X_i$    | 0    | 1    | 2    | 3    | 4    | 5    | 6    |
| $p(x_i)$ | 0.02 | 0.15 | 0.22 | 0.26 | 0.17 | 0.14 | 0.04 |

Find the mean and standard deviation of X.

**CO4****Q 5****20 Marks**

(a) If the probability density function of a continuous random variable X is given by

$$f(x) = \begin{cases} kx^2; & 0 < x < 1 \\ 0; & \text{otherwise} \end{cases}$$

Find the value of  $k$  and hence compute its mean, variance, and standard deviation.

**CO4**

(b) Define the Kurtosis, Skewness, Correlation and Regression analysis with the help of one example. **CO5**

(c) The following are the values of two variables, diastolic blood pressure (y) and time in minutes (x). Diastolic blood pressure (y) was recorded at various time periods (x) on individuals. Find the regression line of y on x and x on y.

|                              |    |    |    |    |    |
|------------------------------|----|----|----|----|----|
| Time periods (x)             | 0  | 5  | 10 | 15 | 20 |
| Diastolic blood pressure (y) | 72 | 67 | 70 | 65 | 66 |

**CO5**

